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Tutorial-3

1. Write a C program that swaps two integer variables using two approaches:
   a) pass by value
   b) pass by reference using pointers.

Ans:

Source Code:

```
#include <stdio.h>

void pass_by_value(int num1,int num2){
 int temp;
 printf("Before swapping (Values in swap function) : \n a = %d b = %d . \n",num1,num2);

 temp=num1;
 num1=num2;
 num2=temp;

 printf("After swapping (Values in swap function): \n a = %d b = %d . \n",num1,num2);
}

void pass_by_reference(int *num1,int *num2){
 int temp;
 printf("Before swapping (Values in swap function) : \n a = %d b = %d . \n",*num1,*num2);

 temp=*num1;
 *num1=*num2;
 *num2=temp;

 printf("After swapping (Values in swap function): \n a = %d b = %d . \n",*num1,*num2);
}

int main(){
 int a=10,b=20;

 printf("===== Swapping by pass by value =====\n \n ");
 printf("Before swapping (Values in main function) : \n a = %d b = %d . \n",a,b);
 pass_by_value(a,b);
 printf("After swapping (Values in main function): \n a = %d b = %d . \n",a,b);

 printf("===== Swapping by pass by reference =====\n \n ");
 printf("Before swapping (Values in main function) : \n a = %d b = %d . \n",a,b);
 pass_by_reference(&a,&b);
 printf("After swapping (Values in main function): \n a = %d b = %d . \n",a,b);

 printf("\n");
 return 0;

}
```

Output :

```
===== Swapping by pass by value =====

 Before swapping (Values in main function) :
 a = 10 b = 20 .
Before swapping (Values in swap function) :
 a = 10 b = 20 .
After swapping (Values in swap function):
 a = 20 b = 10 .
After swapping (Values in main function):
 a = 10 b = 20 .
===== Swapping by pass by reference =====

 Before swapping (Values in main function) :
 a = 10 b = 20 .
Before swapping (Values in swap function) :
 a = 10 b = 20 .
After swapping (Values in swap function):
 a = 20 b = 10 .
After swapping (Values in main function):
 a = 20 b = 10 .
```

2. Write a C program to read n integers from the user, store them in an array, and print the elements using pointers.

Ans:

Source Code:

```c
#include <stdio.h>

void print_by_pointer(int *ptr,int n){
 printf("The numbers you entered are : \n");

 for(int i=0;i<n;i++){
  printf("%d \t",*(ptr+i));
 }
}

int main(){
 int n;

 printf("How many numbers to enter : ");
 scanf("%d",&n);

 int arr[n];
 int *pointer=arr;

 for(int i=0;i<n;i++){
  printf("Enter number no. %d : ",(i+1));
  scanf("%d",(pointer+i));
 }
 print_by_pointer(pointer,n);

 printf("\n");
 return 0;
}
```

Output :

```
How many numbers to enter : 5
Enter number no. 1 : 1325
Enter number no. 2 : 53224
Enter number no. 3 : 144
Enter number no. 4 : 354
Enter number no. 5 : 54634
The numbers you entered are :
1325    53224   144     354     54634
```

3. Write a C program to read n elements into an array and compute the sum of all elements using pointers.

Ans:

Source Code:

```c
#include <stdio.h>

int compute_sum(int *pointer,int n){
 int sum=0;
 for(int i=0;i<n;i++){
  sum=sum+*(pointer+i);
 }
 return sum;
}

int main(){
 int n,result;

 printf("How many integers : ");
 scanf("%d",&n);

 int arr[n];
 int *pointer=arr;

 for(int i=0;i<n;i++){
  printf("Enter number no. %d : ",(i+1));
  scanf("%d",(pointer+i));
 }
 result=compute_sum(pointer,n);
 printf("Sum of the entered numbers : %d.",result);

 printf("\n");
 return 0;
}
```

Output :

```
How many integers : 10
Enter number no. 1 : 1
Enter number no. 2 : 2
Enter number no. 3 : 3
Enter number no. 4 : 4
Enter number no. 5 : 5
Enter number no. 6 : 10
Enter number no. 7 : 9
Enter number no. 8 : 8
Enter number no. 9 : 7
Enter number no. 10 : 6
Sum of the entered numbers : 55.
```

4. Write a C program to read n integers into an array and find the largest element using pointers.

Ans:

Source Code:

```
#include<stdio.h>

int largest_element(int *pointer,int n){
 int largest=*pointer;
 for(int i=1;i<n;i++){
  if(*(pointer+i)>largest){
   largest=*(pointer+i);
  }
 }
 return largest;
}

int main(){
 int n,result;

 printf("How many integers : ");
 scanf("%d",&n);

 int arr[n];
 int *pointer=arr;

 for(int i=0;i<n;i++){
  printf("Enter number no. %d : ",(i+1));
  scanf("%d",(pointer+i));
 }
 result=largest_element(pointer,n);
 printf("Largest element : %d.",result);

 printf("\n");
 return 0;
}
```

Output :

```
How many integers : 8
Enter number no. 1 : 12
Enter number no. 2 : 84
Enter number no. 3 : 92
Enter number no. 4 : 13
Enter number no. 5 : 1
Enter number no. 6 : 46
Enter number no. 7 : 378
Enter number no. 8 : 235
Largest element : 378.
```

5. Write a C program to read n elements into an array and search for a given element using pointers.

Ans:

Source Code:

```c
#include<stdio.h>

int search(int *pointer,int n,int element_to_search){
 for(int i=0;i<n;i++){
  if(*(pointer+i)==element_to_search){
        return i;
   }
  }
 return -1;
}

int main(){
 int n,element_to_search,result;

 printf("How many integers : ");
 scanf("%d",&n);

 int arr[n];
 int *pointer=arr;

 for(int i=0;i<n;i++){
  printf("Enter number no. %d : ",(i+1));
  scanf("%d",(pointer+i));
 }

 printf("What element do you want to search for : ");
 scanf("%d",&element_to_search);

 result=search(pointer,n,element_to_search);
 if (result==-1){
  printf("Element %d does not exist in the array.",element_to_search);
 }else{
 printf("The element %d is at index %d of the array.",element_to_search,result);
 }

 printf("\n");
 return 0;
}
```

Output :

```
How many integers : 5
Enter number no. 1 : 12
Enter number no. 2 : 34
Enter number no. 3 : 65
Enter number no. 4 : 78
Enter number no. 5 : 89
What element do you want to search for : 34
The element 34 is at index 1 of the array.
```

```
How many integers : 4
Enter number no. 1 : 12
Enter number no. 2 : 124
Enter number no. 3 : 53225
Enter number no. 4 : 13325
What element do you want to search for : 253
Element 253 does not exist in the array.
```

6. Write a C program to read n elements into an array and print the array elements in reverse order using pointers.

Ans:

Source Code:

```
#include<stdio.h>

void reverse_print(int *pointer,int n,int size){
 printf("The elements of the array in reverse order : \n");
 for(int i=size-1;i>=0;i--){
   printf("%d\t",*(pointer+i));
  }
}

int main(){
 int n;

 printf("How many integers : ");
 scanf("%d",&n);

 int arr[n];
 int *pointer=arr,size=sizeof(arr)/sizeof(arr[0]);

 for(int i=0;i<n;i++){
  printf("Enter number no. %d : ",(i+1));
  scanf("%d",(pointer+i));
 }

 reverse_print(pointer,n,size);

 printf("\n");
 return 0;
}
```

Output :

```
How many integers : 5
Enter number no. 1 : 143
Enter number no. 2 : 63445423
Enter number no. 3 : 4253
Enter number no. 4 : 3354
Enter number no. 5 : 42344
The elements of the array in reverse order :
42344   3354    4253    63445423        143
```

7. Write a C program to dynamically allocate memory for an integer, a float, and a character at runtime and display their values.

Ans:

Source Code:

```c
#include <stdio.h>
#include <stdlib.h>

int main(){
 int *ptr1;
 float *ptr2;
 char *ptr3;

 ptr1=(int*) malloc(10*sizeof(int));
 ptr2=(float*) malloc(12*sizeof(float));
 ptr3=(char*) malloc(4*sizeof(char));

 printf("Memory allocated to int : %ld",10*sizeof(int));
 printf("\nMemory allocated to float : %ld",12*sizeof(12));
 printf("\nMemory allocated to char : %ld",4*sizeof(char));

 printf("\n");
 return 0;
}
```

Output :

```
Memory allocated to int : 40
Memory allocated to float : 48
Memory allocated to char : 4
```

8. Write a C program that reads an integer n, dynamically allocates memory for n integers, and passes the pointer to a function that returns the sum of the elements.

Ans:

Source Code:

```
#include<stdio.h>
#include<stdlib.h>

int sum_n(int *ptr,int n){
 int sum=0;

 for(int i=0;i<n;i++){
  sum=sum+*(ptr+i);
 }
 return sum;
}

int main(){
 int *ptr,n,sum;

 printf("Enter how many integers : ");
 scanf("%d",&n);

 ptr=(int*) malloc(n*sizeof(int));

 for(int i=0;i<n;i++){
  printf("\nEnter integer no. %d : ",i+1);
  scanf("%d",(ptr+i));
 }

 sum=sum_n(ptr,n);
 printf("Sum = %d",sum);

 printf("\n");
 return 0;
}
```

Output :

```
Enter how many integers : 5

Enter integer no. 1 :
1

Enter integer no. 2 : 2

Enter integer no. 3 : 3

Enter integer no. 4 : 4

Enter integer no. 5 : 5
Sum = 15
```